LEGO Island: Milestone 2 Project Design

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Project Goal

LEGO Island is an interactive data visualization that presents over 50 years of LEGO set history (18,457 records, 1970-2022) as an explorable top-down island map. Taking inspiration from the 1997 video game *LEGO Island*, each geographic district on the map represents a major LEGO theme (Star Wars, City, Duplo, Technic, etc.), with district size proportional to set count. Users can navigate the island by clicking into districts, drilling down through subthemes, and ultimately inspecting individual sets with official box imagery, piece counts, pricing data, and temporal trends. We want to make exploring LEGO's cultural evolution feel immersive and playful while still grounding every interaction in real data.

Visualization Sketches

Sketch 1: Island Overview. This is the landing view. Users see a stylised island with labelled theme districts. Hovering over a district highlights it and shows a tooltip with set count and year range. Clicking zooms into the district detail view.

Instead of a traditional dashboard, users experience LEGO history spatially:

- Larger districts represent themes with more sets
- Clicking a district reveals detailed statistics and set collections
- Users can explore trends such as production over time, piece complexity, and pricing

Figure 1. Top-down island map with theme districts sized by set count, harbour, compass, and tooltip mockup.

Sketch 2: District Drill-Down and Statistics Panel. After clicking a district, users will see a split panel with key statistics of the theme such as total sets created, average pieces, price range, etc. In addition, users will see a scatter plot of sets released per year with a trendline, bar chart of piece count distribution, and a scatter plot of average yearly retail prices. Potentially, we are planning to further analyze theme statistics by having scatter plots of piece count vs. price as shown in Figure 3.

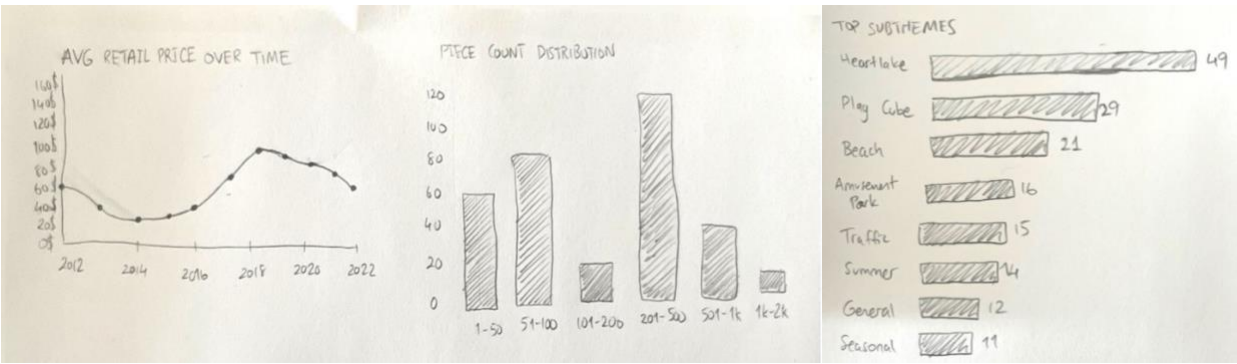

Figure 4. Sketches of planned bar charts and scatter plots to be displayed on theme statistics panel

Sketch 3: Top Subthemes and Featured Sets Panel. Accessible within any district, this section at the bottom of the district panel will display a vertical bar chart of set counts of top subthemes within a theme as shown in sketch below. Next, we will also provide a scrollable grid of featured sets which are popular lego sets in that theme and display their cover image, name, # of pieces and price info displayed in a visually elegant way.

Tools and Relevant Lectures

Component	Tools / Libraries	Relevant Lectures
Island map rendering	D3.js (SVG/Canvas), custom topology generation	L5: Maps, L4: D3

District zoom & navigation	D3-zoom, CSS transitions	L4: Interaction, L7: Perception
Bar charts & timelines	D3.js bar/line chart modules	L3: Tabular data, L4: D3
Scatter plot (price vs. pieces)	D3.js + brushing/linking	L3: Tabular data, L9: Graphs
Set card gallery	HTML/CSS grid, imageURL from dataset	L7: Design, L10: Evaluation
Data pipeline	Python (pandas) for preprocessing, JSON export	L2: Data wrangling
Website framework	Svelte or vanilla JS + Vite bundler	N/A

Core Visualization (Minimum Viable Product)

The MVP delivers the essential drill-down experience in three layers. **(1) Island map with clickable districts:** a static SVG island where at least the 8 largest themes are rendered as labelled, which are clickable regions with logo and road network connecting themes. **(2) District detail view:** clicking a district opens a panel containing a collection of D3 bar/scatter charts of lego set statistics alongside a scrollable grid of set cards (name, thumbnail via imageURL, piece count). Together these two views cover the core data story, including theme scale, temporal trends, and individual theme identity. **(3) Basic statistics:** a scatter plot of sets released per year & avg. retail price over time, bar chart of piece count distribution & top subthemes; all fethched and calculate accurately from dataset and complete with axis labels and a trend line. Missing data will be indicated with a grey 'data unavailable' label rather than imputed.

Extra Ideas

Idea 1) Time slider: a global year slider (1970-2022) that filters the entire island in real time so that districts grow and shrink, new themes appear, and set cards update. This requires efficient data binding and incremental DOM updates.

Idea 2) Subtheme filtering: within a district, users can filter out certain subthemes to analyse data more specifically and go further into subthemes (e.g. Star Wars -> Clone Wars), which will require dyamic data loading.

Idea 3) Comparison mode: users select two themes and see side-by-side view or get data displayed on overlapped plots of price distributions, and piece-count histograms with 2 trendliens for direct cross-theme analysis.

Idea 4) Search: a search bar that lets users find sets by name or ID, jumping directly to the relevant district and highlighting the result. Combined with the time slider, this supports targeted exploration alongside free browsing.

Idea 5) Guided exploration mode: involves a 2D character in a “tour mode” that walks around the map and through key insights listeing keyboard arrows or mouse input to add interactivity to the game. When the character visits a themeland, a short popup box can appear to provide quich theme info. This requires sequencing interactions and overlay annotations.

Idea 6) Artistic and creative elements: incorporate decorative and animated features (e.g. dolphins jumping in and out of water) and add other visual elements like trees, bridges, mini-islands that makes Lego Island experience more immersive and beautiful.

Implementation Breakdown

Task	Owner	Priority	Depends On	Task	Owner	Priority	Depends On
Data cleaning & JSON export	Daniel	Core	None	Zoom & pan transitions	Kian	Extra (E1)	Island layout
Island SVG layout & districts	Kian	Core	Data export	Global time slider	Altu	Extra (E2)	All core charts
District detail panel (bar chart)	Altu	Core	Data export	Subtheme drill-down	Daniel	Extra (E3)	District panel
Set card grid + thumbnails	Daniel	Core	Data export	Comparison mode	Altu	Extra (E4)	Statistics panel
Scatter plot (pieces vs. price)	Altu	Core	Data export	Set detail modal	Daniel	Extra (E5)	Set card grid
Breadcrumb navigation	Kian	Core	Island + District	Search bar	Kian	Extra (E6)	All core views